SKYFI MCP

Ian Zepp

March 2026

ARCHITECTURE

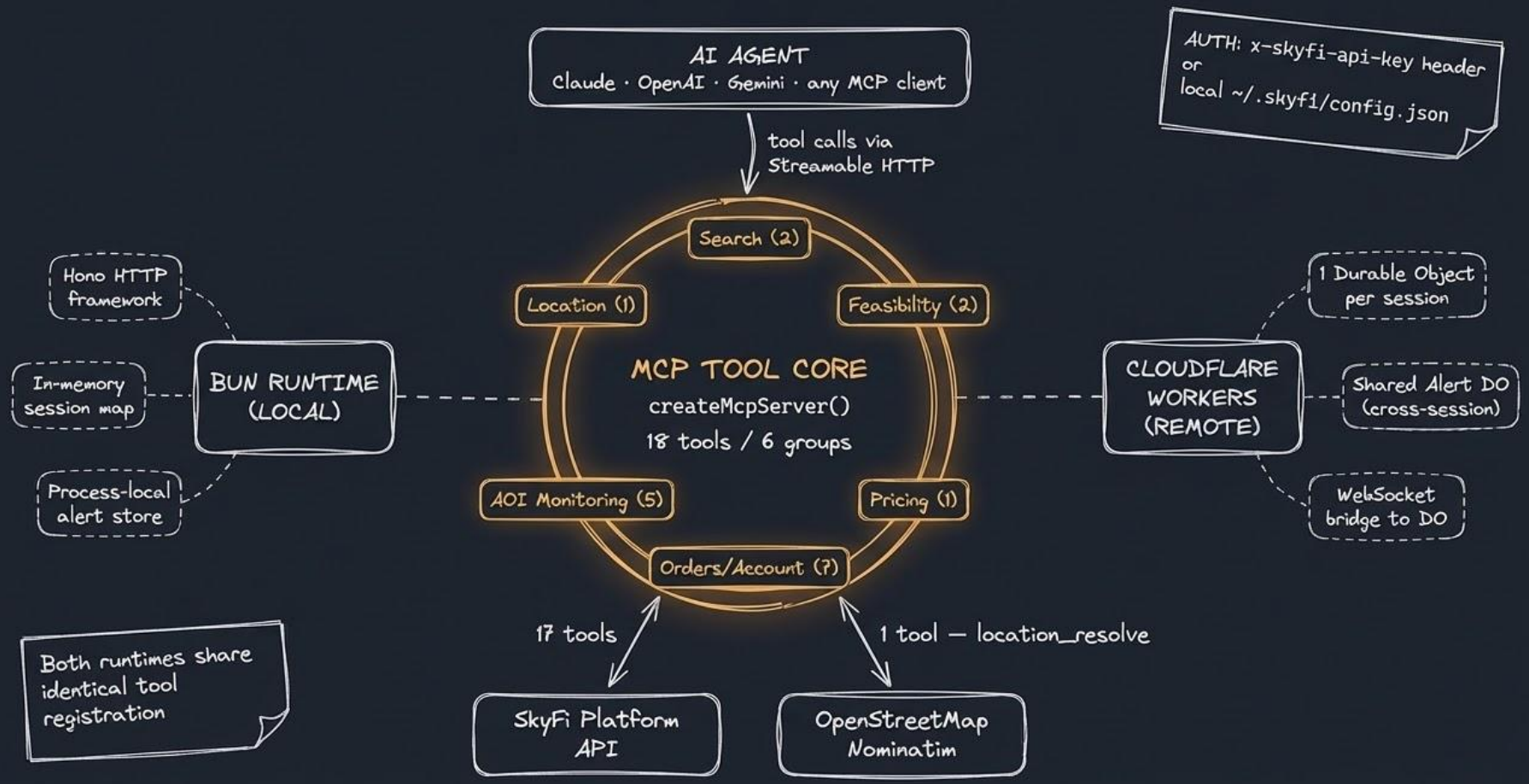
CONFIRMATION GATE

EVAL HARNESS

COVERAGE

CLI

SYSTEM ARCHITECTURE



CONFIRMATION GATE

1. **orders_prepare** → validate params, fetch pricing, store token ← `crypto.randomUUID()` — *in-memory or Durable Object*
2. Return pricing + confirmation token to agent — **ORDER NOT YET PLACED**
3. **HUMAN REVIEW** — agent presents pricing, user decides
   - *Configurable TTL (default 5 min)*
   - *— single use*
   - *— no bypass*
   - 04:59
   - *Token expires → restart from Step 1*
4. **orders_confirm** → await atomic consume (get + delete)
   - *token invalidated immediately* → *Double submit → already consumed → rejected*
5. **Submit to SkyFi API** → return order ID — **ORDER EXECUTED**
   - *API error → restore() puts token back → retry*

LIVE PROOF

- *Deep research skill exercises full gate*
- *Pinned zero-cost open-data archive*
- ***deliveryDriver: NONE** (no cloud creds needed)*
- ***Eval:** live approval + expired-token scenarios*
- ***Token store:** Durable Object on Cloudflare — cross-session persistence*

*The token is the only path from pricing to purchase — **the agent cannot fabricate, reuse, or skip it***

EVAL HARNESS — AGENT BEHAVIOR VALIDATION

SCENARIO INPUT

- human-approval [ordering]
- archive-search [search]
- feasibility-chain [tasking]
- aoi-lifecycle [monitoring]
- no-confirm [safety]
- live-search [live]

37 scenarios across 15 suites

one at a time

EXECUTION LOOP

Initial prompt + allowed tools → Model → Tool Call → Execute → Result → Model

up to 8 steps

FIXTURE: canned YAML responses

LIVE: real MCP → SkyFi API

final answer + tool call trace

GRADING

DETERMINISTIC GRADER

- ☑ Expected tools called
- ☒ Tool sequence order
- ☒ Forbidden tools avoided
- ☑ Must-contain phrases
- ☑ Any-of phrases
- ☒ Must-not-contain
- ☑ Min answer length
- ☒ Tool output content

PASS or FAIL + reasons

SECONDARY JUDGE

Claude Sonnet 4.5 (OpenRouter)
Only invoked on failures

→ real_failure
→ rubric_too_strict
→ ambiguous

Separates actual bugs from overly strict rubrics

Also: 137 unit tests / 21 test files (function-level, separate from evals)

REQUIREMENTS COVERAGE

Transport: Streamable HTTP – intentional upgrade from legacy SSE

Cloudflare: 3 Durable Objects – MCP session, AlertStore, ConfirmationStore

- Remote MCP – Cloudflare Workers + Durable Objects
- SkyFi API – 18 tools across 6 groups
- Local Hosting – Bun runtime
- Conversational Ordering – prepare/confirm gate
- Feasibility + Pass Prediction
- AOI Monitoring – webhooks + alert persistence
- Auth – header-based (cloud) + JSON config (local)
- Integration Docs – 7 platforms (ADK, LangChain, AI SDK, Claude Web, OpenAI, Claude Code, Gemini)
- Demo Agent – geospatial deep-research skill

Payments: billing through SkyFi account – no separate payment subsystem needed

10 / 10 REQUIREMENTS MET

SKYFI-CLI // NATIVE ALTERNATIVE

Same SkyFi API, same workflows, two interfaces — pick the right one for the job

MCP SERVER (TypeScript)

AI Agent

tool calls via Streamable HTTP

Session state in Durable Objects

MCP Server (18 tools)

Rich agent autonomy
Multi-step tool loops

Best for: conversational workflows

SkyFi API

Token cost: full MCP overhead per call

CLI (Rust, 5.5K LOC)

Terminal / Script

direct commands

Zero token overhead

skyfi-cli binary

Structured JSON output for piping

```
├─ archives search | get
├─ orders list | get | order-archive | order-tasking
│        | pass-targeted | download
├─ feasibility check | status | pass-predictions
├─ notifications create | list | get | delete
├─ alerts poll | watch | install
├─ pricing
└─ research
```

Homebrew installable

Best for: scripting, automation, low-cost exploration

SkyFi API

ALERTS INSTALL — OS-NATIVE NOTIFICATIONS

skyfi-cli alerts install

macOS: launchd agent
→ Notification Center banners via osascript

Linux: systemd user service + timer
→ notify-send desktop notifications

optional

optional

→ --on-alert hook:
spawns user script per alert, JSON on stdin + env vars

research command: prompt-driven agent loop, writes markdown briefs + JSON trace

Polling interval configurable. Seen/unseen state persisted locally. No MCP server required.

Live Demo